

Q1) 4

assume $K=N$,

Q1) $D=4096$, $F=40$, $V=32000$, $L=64$, so we have

$$L(3DF + 2D(N+K)H) + DV \text{ total}$$

$$= L(3DF + 2D^2 + 2DK(2DN + 2DK)H) + DV \text{ total}$$

$$= L(3DF + 4D^2 + 4DK^2) + DV$$

$$\approx 17.3 \text{ billion parameters}$$

$$\text{Fraction of which are attention} = \frac{(64)(40^2)}{17.3B} = 25\%$$

KV Cache size per token = $2SLKH$ in total, with $2LKH$ per token $\approx 524 \text{ KB}$ ~~$\approx 524 \text{ MB}$~~

Q2) To compute $A[B,D] * W[D,F]$, we do $2BDF$ FLOPs.

But we are sharded over X and Y but not Z , so we have

4 replicas of our matrix. Therefore, we would compute $8BDF$ FLOPs.

We have 128 devices in total, so each TPU does $\frac{8BDF}{16}$ FLOPs.

* For the given mesh $\{X=4, Y=8, Z=4\}$

$$Q3) A[I, S, K, L] * B[I, S, M, N, O] \rightarrow C[K, L, M, N, O]$$

▷ We are contracting over I, S and batching over nothing! So, it takes $2ISKLMNO$ flops.

Q4) Arithmetic Intensity of Self Attention (in terms of K, T, S , and N only projections).

Self Attention takes $\approx \frac{4}{3} K^2 T N H$ flops (assume no cache)
 and it looks like $\frac{4}{3}$ because it is from scratch computation

$BTKQH$ ~~$BTKQH$~~
 + $2BSKH$ if we assume
 KG reshapes to
 $N=K$, then
 $S=T$

$$\begin{aligned} & BTKH \\ & + 2BSKH \\ & \hline & 3BSKH \end{aligned}$$

$$\text{So, arithmetic intensity} = \frac{4T}{3BSKH} = \frac{4T}{3} \text{ if } K=N$$

$$\text{with an inner product} = \frac{4T}{3} \text{ ie } 4T \text{ flops}$$

ie byte level.

So, ~~the~~ assuming Q, K, V reductions are all $2x2x8$, we have
 ~~compute~~ Intensity = $\frac{4T}{3}$ ie we are compute some approx when $T > 225$ (no cache)

When $T < 225$, t_{comm} dominates attention.

FFW takes $\frac{2}{3DF}$ FLOPs ie $2BT$ intensity. It's
per $3DF$ bits compute time assuming $B=8$,
when $T \geq 225$

$$4th \text{ comm} = 8BSKH$$

$$FFW \text{ comm} = 3DF + 2STD + 2STF$$

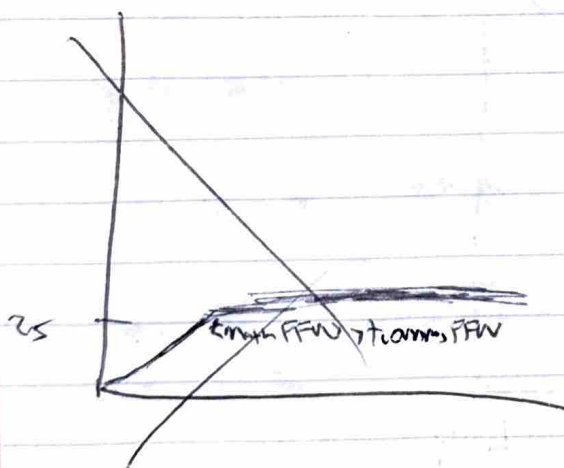
assuming $F=40$ and $77B$, FFW comm clearly dominates
and $KH=0$

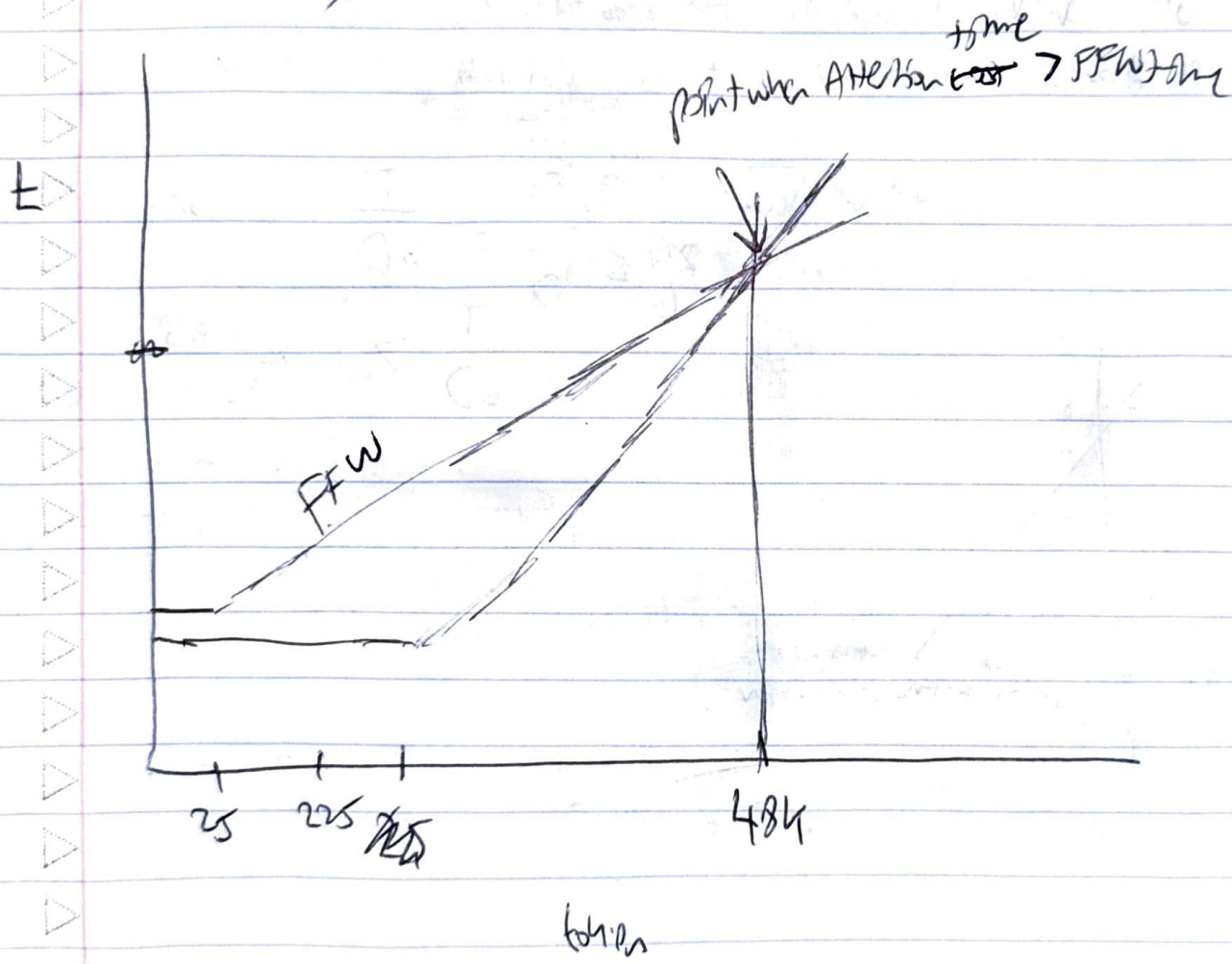
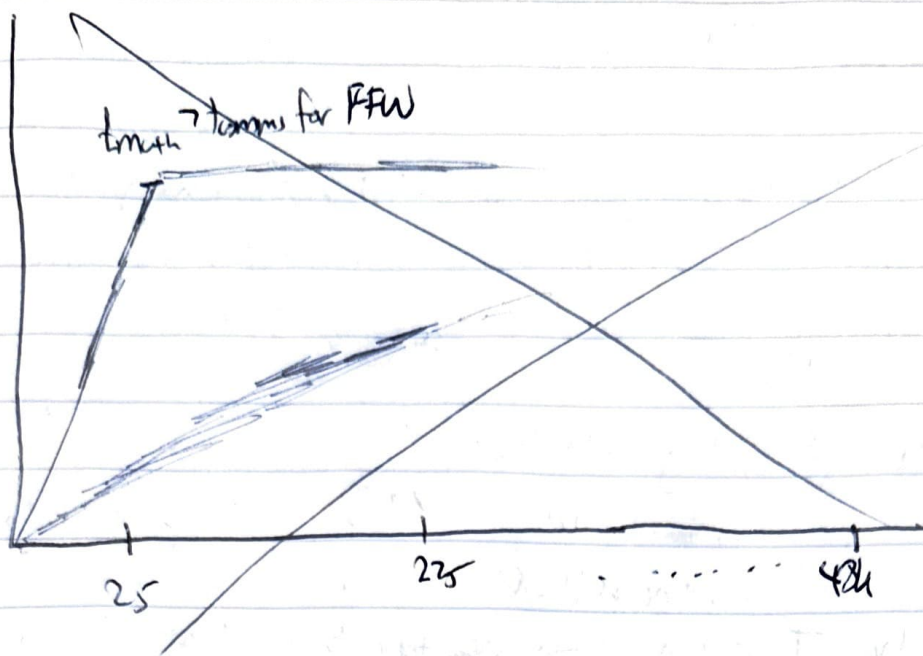
So, when $T < 225$, FFW ~~is~~ takes longer. When $T \geq 225$,
attention is compute bound and takes $4ST^2/NH$ while FFW takes
 $6BSTDF$.

$$\text{so we have } \frac{4ST^2}{6} = \frac{T}{60} \text{ ie when}$$

$$\frac{T}{60} > 1, \text{ attention takes more time}$$

ie when $T > 60$
or $T > 48,000$





$$(a5) \text{ Self-Attention FLOPs} = \frac{4}{1} B T^3 N H = 4 B T^3 N H$$

$$\text{Projection FLOPs} = \frac{4}{1} B T D H (N+K) = 4 B T D H (N+K)$$

$$4 B T^2 N H = 4 B T D H (N+K) \Rightarrow T D = D N + D K \text{ ie}$$

$$\boxed{T = 2D}$$

(a6) Say we only save out of 7 matmuls. (0, 4, 1, 0 + FFW). How many extra FLOPs do we need to rematerialize in the backward pass?

A) We need to reinit dot product attention + softmax + calculate values + residual + norm_p + gelu + element wise * + norm_p.

$$\sigma(A_{n1}) [B, T, F] * A_{n2} [B, T, F] = O(BTF)$$

~~dot product attention is~~

layer norm is $O(BTD)$, residual, are $O(BTD)$

~~This is mostly dominated by attention, ie~~

$$\text{Attention cost is } [B T K G H] \cdot H \cdot [B S K H] = 2 B T S K G H, \text{ assume } T = S, \text{ so we have } 2 B T^2 K G H$$

$$\text{also, } [B T S K G] [B S K H] = 2 B T S K G H = 2 B T^2 K G H, \text{ so total} \\ = \boxed{4 B T^2 K G H \text{ FLOPs remat}}$$

$$24571 \leq BT \leq \frac{1}{2K}$$

Q7) FP8 FLOPs, no structured sparsity. We know that roughly
 $6 * \text{tokens} * \text{activated params} = \text{training FLOPs}$.

activated params = ~~37e9~~ 37e9

tokens = 14.8e12 (assume 1 epoch of training)

a priori estimate = $6 * 37e9 * 14.8e12 = 3.28e24$ training FLOPs

actual use = 2.79M H800 hours. This corresponds to An H800

achieves 3,026 FP8 TFLOPs with structured sparsity, without which

it achieves ≈ 1513 FP8 TFLOPs. So total, ^{theoretical} FLOPs the kernel for =

$2.79e6 * 60 * 60 * 1513e12 = 1.520e25$ theoretical FLOPs.

Hardware Utilization = $\frac{\text{realized}}{\text{theoretical}} = \frac{3.28e24}{1.520e25} = \boxed{21.6\%}$

Q8) Since we want to overlap operations, we probably want to load all
 E exps into memory. Also assume its all on one TPV so is just
 local from HBM.

~~$t_{\text{main}} = \text{BTD} + 3\text{DFE}$~~ We do K BTD & DF matrix multiplying
 ie we have $t_{\text{main}} = \text{BTD} + 3\text{DFE} / \text{accel FLOPs/s}$
 and $t_{\text{comm}} = \text{BTD} + 3\text{DFE} / \text{accel Bandwidth}$
~~BTF~~

~~Intensity (MB/s) = $K(6\text{BTD})$~~ If we assume $B \leq 2D < F = 4D$, the
 ~~$\text{BTD} + 3\text{BTF} + 3\text{DF}$~~
~~this is $K(6\text{BTD})$~~ ~~$2K\text{BTD}$~~ ~~We are, so~~

Intensity (MOE) = $\frac{k(6BTDF)}{BSF + BTDF + IDFE}$, if we assume B small, this is

$$\frac{k(6BTDF)}{IDFE} = \frac{2kBT}{E}$$

~~To be complete band, we want $\frac{2kBT}{E} > 1 \Leftrightarrow BT > \frac{E}{2k}$~~

~~$E=256, k=8$~~ , and use with $\frac{1.97214}{8-2e11}$ intensity, we have

~~that DeepJank has complete band when~~, therefore

we are complete band when $\frac{2kBT}{E} > \frac{1.97214}{8-2e11} \Leftrightarrow$

$$\frac{2kBT}{E} > 240 \Leftrightarrow BT > \frac{120E}{k}$$

So for DeepJank with $k=8, E=256$, we need $BT > \frac{(120)(256)}{8} = \underline{3840}$